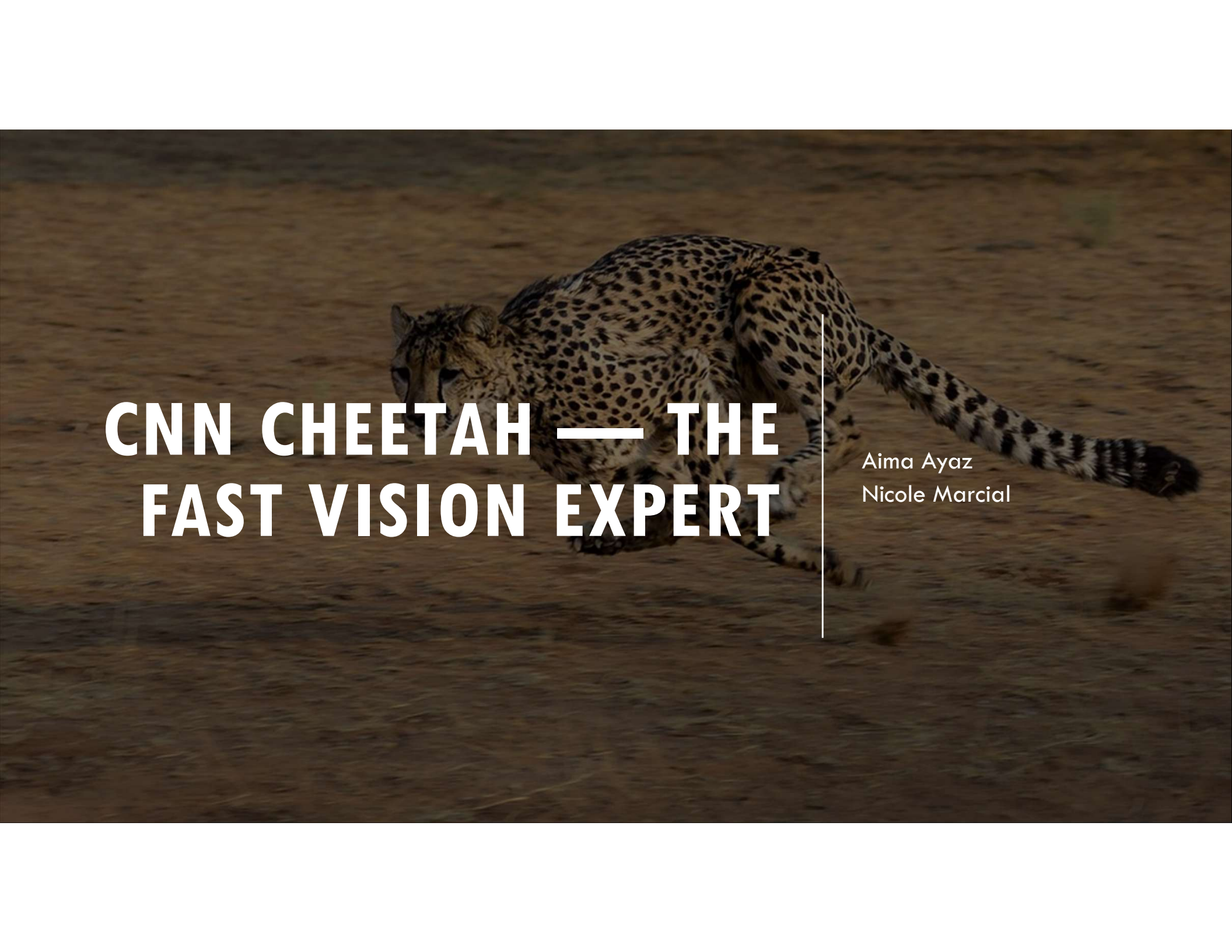


A cheetah is captured in mid-stride, running across a dry, brown, and textured landscape. The cheetah's body is low to the ground, and its legs are extended, conveying a sense of speed and motion. The background is a blurred expanse of the same dry terrain, emphasizing the cheetah's movement.

CNN CHEETAH — THE FAST VISION EXPERT

Aima Ayaz
Nicole Marcial

INTRODUCTION TO NEURAL NETWORKS

Neural networks are inspired by the brain.

Each neuron: inputs $\rightarrow$ weights $\rightarrow$ bias $\rightarrow$ activation $\rightarrow$ output.

Layers of neurons form a network that learns from data.

Deep learning = many layers that learn features automatically.

ARTIFICIAL
NEURAL NETWORK



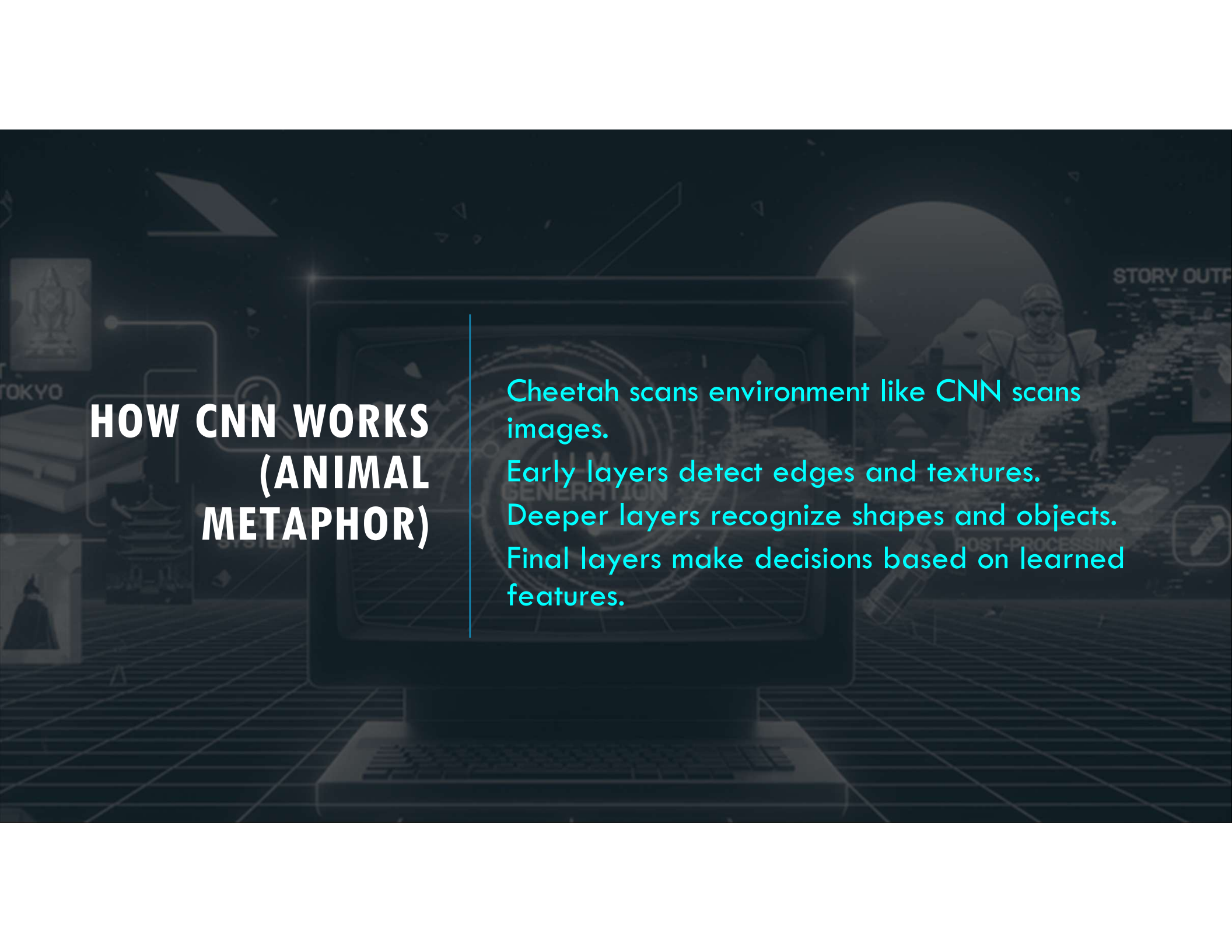
CNN CHEETAH CONCEPT

Animal: Cheetah

Network Type: Convolutional
Neural Network (CNN)

Fast at processing images, sharp
vision, efficient pattern detector.

CNNs are the cheetahs of the
neural network zoo.



HOW CNN WORKS (ANIMAL METAPHOR)

Cheetah scans environment like CNN scans images.

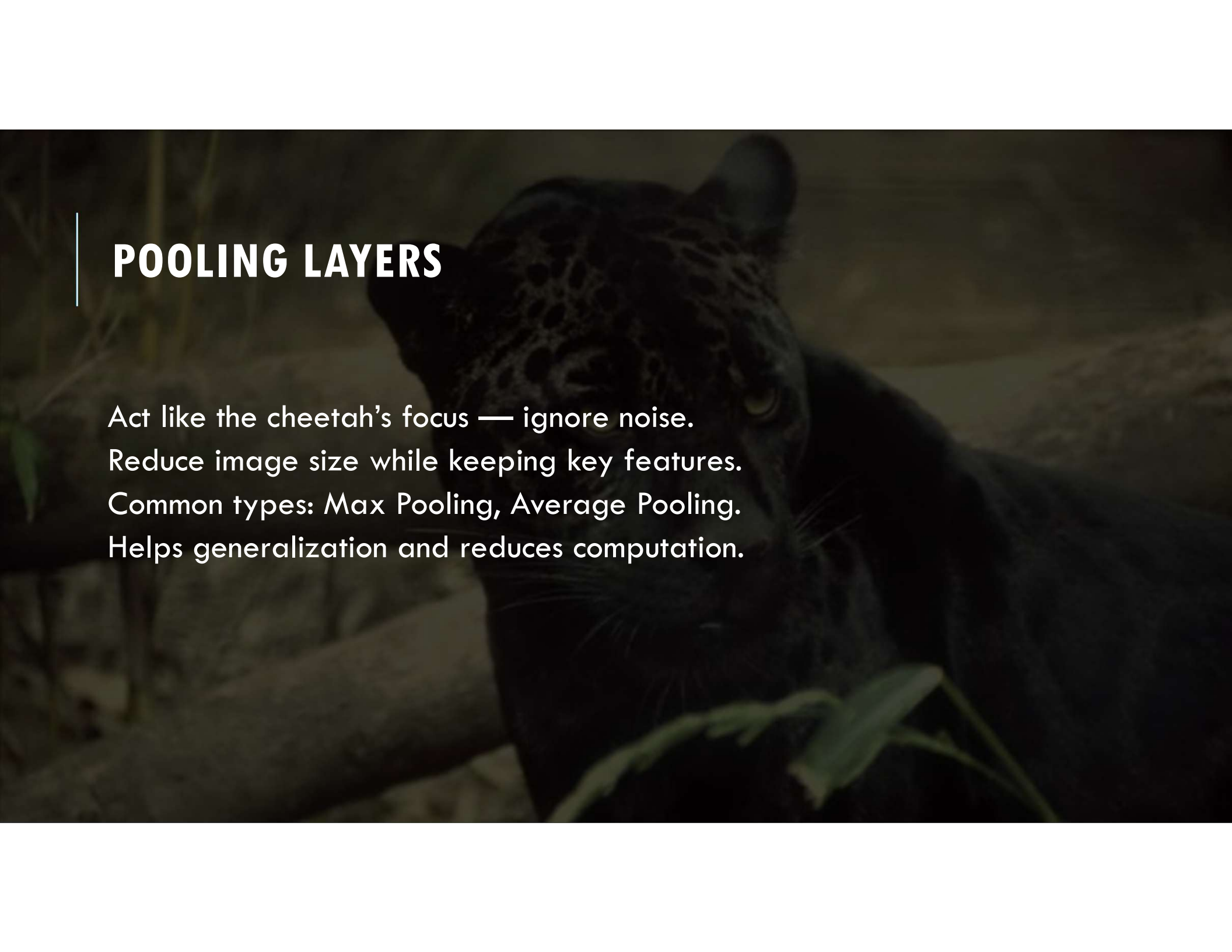
Early layers detect edges and textures.

Deeper layers recognize shapes and objects.

Final layers make decisions based on learned features.

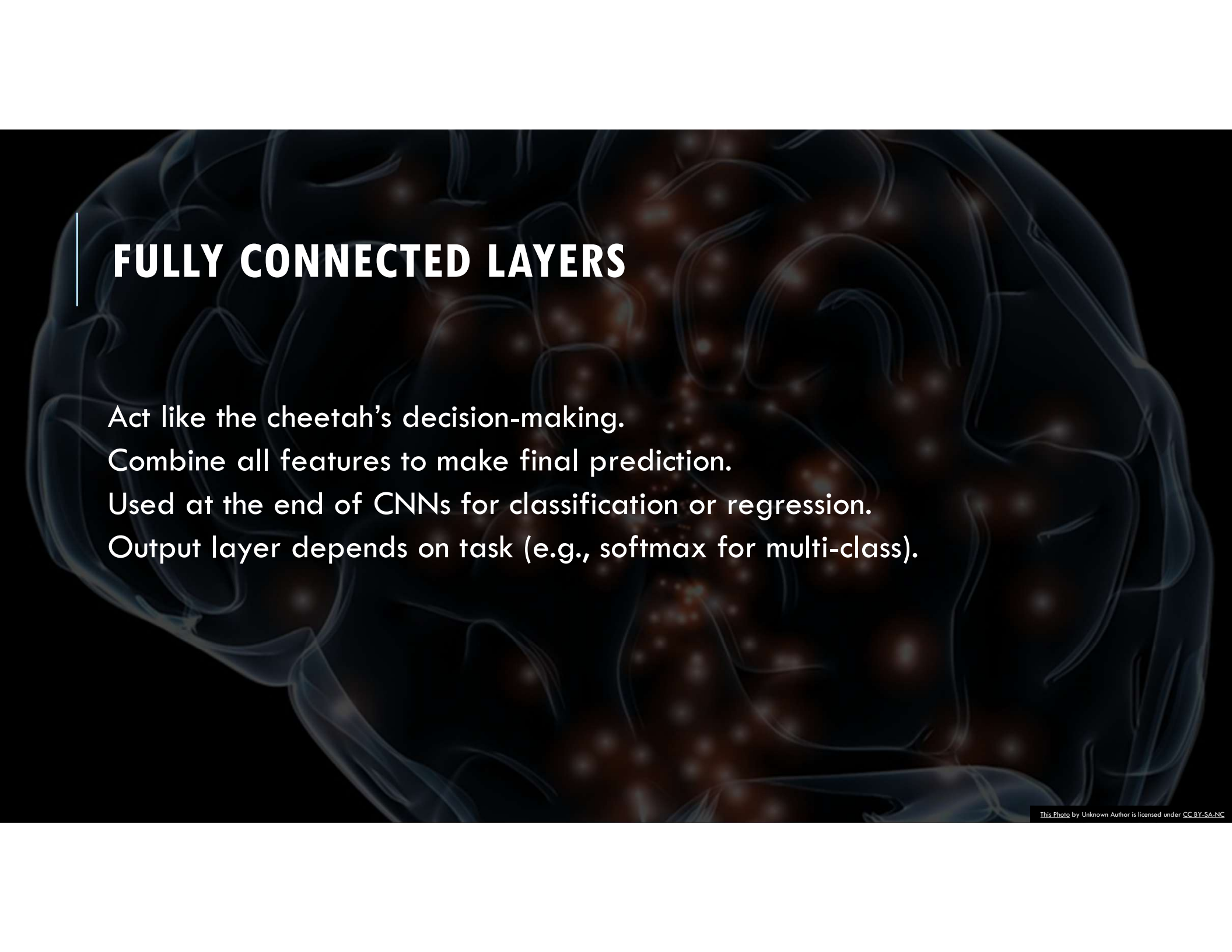
CONVOLUTION LAYERS

**Act like the cheetah's eyes — detect patterns.
Use filters to scan for edges, corners, textures.
Each filter creates a feature map.
Early layers: simple patterns. Deeper: complex features.**



POOLING LAYERS

Act like the cheetah's focus — ignore noise.
Reduce image size while keeping key features.
Common types: Max Pooling, Average Pooling.
Helps generalization and reduces computation.



FULLY CONNECTED LAYERS

Act like the cheetah's decision-making.

Combine all features to make final prediction.

Used at the end of CNNs for classification or regression.

Output layer depends on task (e.g., softmax for multi-class).

APPLICATIONS OF CNNs

Image classification (e.g., cats vs dogs).

Object detection (e.g., self-driving cars).

Medical imaging (e.g., tumor detection).

Face recognition, OCR, quality inspection.





NEURAL NETWORK ZOO — ANIMAL COMPARISON

Animal	Network Type	Specialty
CNN Cheetah	Convolutional Neural Network	Vision, image recognition
RNN Raccoon	Recurrent Neural Network	Sequences, time series
LSTM Lemur	Long Short-Term Memory	Long-term dependencies
Transformer Owl	Transformer	Attention, language modeling
GAN Chameleon	Generative Adversarial Network	Image generation, mimicry
Autoencoder Beaver	Autoencoder	Compression, reconstruction

REFLECTION

The CNN Cheetah helped us understand how CNNs process visual data. Like a cheetah, CNNs scan and detect patterns efficiently. We learned how layers work together to extract features. This metaphor made deep learning concepts easier to grasp.